Final Writeup

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```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```r
library(stringr)
library(ggplot2)
library(readr)
library(janitor)
```

```
## Warning: package 'janitor' was built under R version 4.3.3

##
## Attaching package: 'janitor'

## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

Introduction

Pollinators are essential for maintaining global biodiversity and food security. However, climate change is introducing a critical stressor to these systems: extreme heat. bees rely on environmental temperature to regulate their metabolism, operating within specific "thermal windows" where flight is efficient (Corbet et al.). When temperatures exceed a species' limit, they must cease foraging to avoid overheating.

Crucially, thermal tolerance varies by species. Evidence suggests that introduced European Honey Bees may possess wider thermal windows than many native bees adapted to temperate climates (Abou-Shaara et al.). Additionally, rising temperatures increase Vapor Pressure Deficit (VPD), or the "drying power" of the air. High VPD accelerates water loss, posing a severe desiccation risk to native bees compared to robust introduced species.

While honey bee responses to heat are well-documented (Brovarskiy), there is a significant gap in understanding how diverse native bee communities respond to these pressures in the field. This study bridges that gap by coupling high-resolution climate data with community science records from the Oregon Bee Atlas to answer two questions:

Does extreme heat differentially impact foraging abundance? I predict native bee activity will decline sharply at high temperatures, while introduced species will remain resilient.

Is foraging limited by air dryness (VPD)? I predict that high VPD will negatively correlate with activity, disproportionately affecting native species.

Data

Data Sets Used:

Oregon Bee Atlas (OBA): An open source dataset containing foraging records of bee species in Oregon.

PRISM Climate Group Data: Daily meteorological time-series data obtained from the PRISM Climate Group (Oregon State University). Variables included daily Minimum Temperature (tmin), Mean Temperature (tmean), Maximum Temperature (tmax), Minimum Vapor Pressure Deficit (vpdmin), and Maximum Vapor Pressure Deficit (vpdmax).

Data Skills Demonstrated:

Data Aggregation & Summarization: Compiling multiple weather station files and calculating regional daily averages to prevent record duplication during joining.

Relational Joining: Performing a left-join operation to merge biological occurrence data with meteorological time-series data based on a common Date key.

Data Tidying & Cleaning: Standardizing column names, parsing date formats (mdy vs ymd), and filtering for the active summer season (May–September) to remove seasonal bias.

```r
df <- read.csv("data/OBA_2018-2024.csv")
head(df)
```

```
##   fieldNumber catalogNumber                                occurrenceID userId
## 1     1800001               https://osac.oregonstate.edu/OBS/OBA_1800001 429964
## 2     1800002               https://osac.oregonstate.edu/OBS/OBA_1800002 429964
## 3     1800003               https://osac.oregonstate.edu/OBS/OBA_1800003 429964
## 4     1800004               https://osac.oregonstate.edu/OBS/OBA_1800004 429964
## 5     1800005               https://osac.oregonstate.edu/OBS/OBA_1800005 429964
## 6     1800006               https://osac.oregonstate.edu/OBS/OBA_1800006 429964
##        userLogin firstName firstNameInitial      lastName           recordedBy
## 1 amelathopoulos    Andony               A. Melathopoulos Andony Melathopoulos
```

```
## 2 amelathopoulos    Andony              A. Melathopoulos Andony Melathopoulos
## 3 amelathopoulos    Andony              A. Melathopoulos Andony Melathopoulos
## 4 amelathopoulos    Andony              A. Melathopoulos Andony Melathopoulos
## 5 amelathopoulos    Andony              A. Melathopoulos Andony Melathopoulos
## 6 amelathopoulos    Andony              A. Melathopoulos Andony Melathopoulos
##   sampleId specimenId day month year verbatimEventDate day2 month2 year2
## 1                 NA   4     5 2018            5/4/18   NA           NA
## 2                 NA   4     5 2018            5/4/18   NA           NA
## 3                 NA   4     5 2018            5/4/18   NA           NA
## 4                 NA   4     5 2018            5/4/18   NA           NA
## 5                 NA  16     5 2018           5/16/18   NA           NA
## 6                 NA  22     6 2018           6/22/18   NA           NA
##   startDayofYear endDayofYear country stateProvince  county    locality
## 1             NA                  USA            OR  Benton   Corvallis
## 2             NA                  USA            OR  Benton   Corvallis
## 3             NA                  USA            OR  Benton   Corvallis
## 4             NA                  USA            OR  Benton   Corvallis
## 5             NA                  USA            OR   Wasco      Rowena
## 6             NA                  USA            OR Klamath Modoc Point
##   verbatimElevation decimalLatitude decimalLongitude
## 1                71         44.5599        -123.2883
## 2                71         44.5599        -123.2883
## 3                71         44.5599        -123.2883
## 4                71         44.5599        -123.2883
## 5                62         45.6684        -121.2457
## 6              1271         42.4885        -121.8769
##   coordinateUncertaintyInMeters samplingProtocol relationshipOfResource
## 1                            NA       aerial net                     NA
## 2                            NA       aerial net                     NA
## 3                            NA       aerial net                     NA
## 4                            NA       aerial net                     NA
## 5                            NA       aerial net                     NA
## 6                            NA       aerial net                     NA
##                                   resourceID
## 1 https://osac.oregonstate.edu/OBS/OBA_1800001
## 2 https://osac.oregonstate.edu/OBS/OBA_1800002
## 3 https://osac.oregonstate.edu/OBS/OBA_1800003
## 4 https://osac.oregonstate.edu/OBS/OBA_1800004
## 5 https://osac.oregonstate.edu/OBS/OBA_1800005
## 6 https://osac.oregonstate.edu/OBS/OBA_1800006
##                      relatedResourceID relationshipRemarks  phylumPlant
## 1 1444041d-1464-49d0-9c5c-f297aa90e2ae                  NA Tracheophyta
## 2 1444041d-1464-49d0-9c5c-f297aa90e2ae                  NA Tracheophyta
## 3 1444041d-1464-49d0-9c5c-f297aa90e2ae                  NA Tracheophyta
## 4 1444041d-1464-49d0-9c5c-f297aa90e2ae                  NA Tracheophyta
## 5 a03202bf-4808-47fe-a57a-d2541249443d                  NA Tracheophyta
## 6 b423e04b-db31-4c6e-abe1-b321e1d16407                  NA Tracheophyta
##     orderPlant   familyPlant   genusPlant            speciesPlant
## 1 Ranunculales Berberidaceae
## 2 Ranunculales Berberidaceae
## 3 Ranunculales Berberidaceae
## 4 Ranunculales Berberidaceae
## 5      Fabales      Fabaceae        Vicia           Vicia villosa
## 6 Ranunculales  Papaveraceae Eschscholzia Eschscholzia californica
```

```
##   taxonRankPlant                                               url phylum class
## 1         family https://www.inaturalist.org/observations/12167479     NA    NA
## 2         family https://www.inaturalist.org/observations/12167479     NA    NA
## 3         family https://www.inaturalist.org/observations/12167479     NA    NA
## 4         family https://www.inaturalist.org/observations/12167479     NA    NA
## 5        species https://www.inaturalist.org/observations/12538376     NA    NA
## 6        species https://www.inaturalist.org/observations/13685289     NA    NA
##   order family   genus subgenus specificEpithet taxonomicNotes scientificName
## 1    NA     NA Andrena                 prunorum
## 2    NA     NA Andrena
## 3    NA     NA Andrena                 hippotes
## 4    NA     NA Andrena           angustitarsata
## 5    NA     NA  Bombus              vosnesenskii
## 6    NA     NA  Bombus              vosnesenskii
##      sex  caste taxonRank identifiedBy familyVolDet genusVolDet
## 1   male               NA     L.R.Best
## 2   male               NA     L.R.Best
## 3   male               NA   A.S.Jackson
## 4 female               NA   A.S.Jackson
## 5 female worker        NA     L.R.Best
## 6 female worker        NA     L.R.Best
##   specificEpithetVolDet sexVolDet casteVolDet
## 1
## 2
## 3
## 4
## 5
## 6
```

```
bee_df <- df %>%
  mutate(date_parsed = mdy(verbatimEventDate)) %>%
  arrange(date_parsed)
```

```
## Warning: There was 1 warning in ‘mutate()‘.
## i In argument: ‘date_parsed = mdy(verbatimEventDate)‘.
## Caused by warning:
## !  147 failed to parse.
```

```
head(bee_df)
```

```
##             fieldNumber catalogNumber occurrenceID userId userLogin firstName
## 1 Jerry_Paul:18.007.001                            175529 jerrypaul     Jerry
## 2 Jerry_Paul:18.007.002                            175529 jerrypaul     Jerry
## 3 Jerry_Paul:18.007.003                            175529 jerrypaul     Jerry
## 4 Jerry_Paul:18.007.004                            175529 jerrypaul     Jerry
## 5 Jerry_Paul:18.007.005                            175529 jerrypaul     Jerry
## 6 Jerry_Paul:18.007.006                            175529 jerrypaul     Jerry
##   firstNameInitial lastName recordedBy sampleId specimenId day month year
## 1               J.     Paul Jerry Paul        7          1  15     3 2017
## 2               J.     Paul Jerry Paul        7          2  15     3 2017
## 3               J.     Paul Jerry Paul        7          3  15     3 2017
## 4               J.     Paul Jerry Paul        7          4  15     3 2017
## 5               J.     Paul Jerry Paul        7          5  15     3 2017
```

```
## 6              J.     Paul Jerry Paul        7          6  15     3 2017
##   verbatimEventDate day2 month2 year2 startDayofYear endDayofYear country
## 1          3/15/17   NA          NA             NA                 USA
## 2          3/15/17   NA          NA             NA                 USA
## 3          3/15/17   NA          NA             NA                 USA
## 4          3/15/17   NA          NA             NA                 USA
## 5          3/15/17   NA          NA             NA                 USA
## 6          3/15/17   NA          NA             NA                 USA
##   stateProvince county                   locality verbatimElevation
## 1            OR Benton Corvallis, Highland Dell Dr               131
## 2            OR Benton Corvallis, Highland Dell Dr               131
## 3            OR Benton Corvallis, Highland Dell Dr               131
## 4            OR Benton Corvallis, Highland Dell Dr               131
## 5            OR Benton Corvallis, Highland Dell Dr               131
## 6            OR Benton Corvallis, Highland Dell Dr               131
##   decimalLatitude decimalLongitude coordinateUncertaintyInMeters
## 1          44.603         -123.264                            NA
## 2          44.603         -123.264                            NA
## 3          44.603         -123.264                            NA
## 4          44.603         -123.264                            NA
## 5          44.603         -123.264                            NA
## 6          44.603         -123.264                            NA
##   samplingProtocol relationshipOfResource resourceID relatedResourceID
## 1       aerial net                    NA
## 2       aerial net                    NA
## 3       aerial net                    NA
## 4       aerial net                    NA
## 5       aerial net                    NA
## 6       aerial net                    NA
##   relationshipRemarks phylumPlant orderPlant familyPlant genusPlant
## 1                  NA
## 2                  NA
## 3                  NA
## 4                  NA
## 5                  NA
## 6                  NA
##   speciesPlant taxonRankPlant url phylum class order family genus subgenus
## 1                                  NA    NA    NA     NA
## 2                                  NA    NA    NA     NA Osmia
## 3                                  NA    NA    NA     NA
## 4                                  NA    NA    NA     NA
## 5                                  NA    NA    NA     NA
## 6                                  NA    NA    NA     NA
##   specificEpithet taxonomicNotes scientificName  sex caste taxonRank
## 1                                                                 NA
## 2        lignaria                               male              NA
## 3                                                                 NA
## 4                                                                 NA
## 5                                                                 NA
## 6                                                                 NA
##   identifiedBy familyVolDet genusVolDet specificEpithetVolDet sexVolDet
## 1
## 2     L.R.Best
## 3
```

```
## 4
## 5
## 6
##   casteVolDet date_parsed
## 1              2017-03-15
## 2              2017-03-15
## 3              2017-03-15
## 4              2017-03-15
## 5              2017-03-15
## 6              2017-03-15
```

```
weather_files <- list.files(path = "data", pattern = "^PRISM.*\\.csv$", full.names = TRUE)

df_list <- list()

if (length(weather_files) > 0) {
  for (i in seq_along(weather_files)) {
    df_list[[i]] <- read_csv(weather_files[i], show_col_types = FALSE) %>%
      clean_names() %>%
      mutate(source_file = basename(weather_files[i]))
  }
} else {
  stop("No files found matching the pattern inside the 'data' folder. Check your directory structure.")
}
weather_df <- bind_rows(df_list)

print(head(weather_df))
```

```
## # A tibble: 6 x 7
##   date       tmin_degrees_f tmean_degrees_f tmax_degrees_f vpdmin_h_pa
##   <date>              <dbl>           <dbl>          <dbl>       <dbl>
## 1 2018-05-04           43.1            58             72.9        0.15
## 2 2018-05-05           48.2            56.8           65.5        0.71
## 3 2018-05-06           47.8            60.2           72.7        0.42
## 4 2018-05-07           48.6            57.1           65.6        0.53
## 5 2018-05-08           45.4            58.5           71.5        0.22
## 6 2018-05-09           46.5            60.1           73.7        0.22
## # i 2 more variables: vpdmax_h_pa <dbl>, source_file <chr>
```

```
temp_df_agg <- weather_df %>%
  group_by(date) %>%
  summarize(
    tmin_degrees_f  = mean(tmin_degrees_f, na.rm = TRUE),
    tmean_degrees_f = mean(tmean_degrees_f, na.rm = TRUE),
    tmax_degrees_f  = mean(tmax_degrees_f, na.rm = TRUE),
    vpdmin          = mean(vpdmin_h_pa, na.rm = TRUE),
    vpdmax          = mean(vpdmax_h_pa, na.rm = TRUE)
  ) %>%
  mutate(date_common = ymd(date))

bee_df_ready <- bee_df %>%
  mutate(date_common = mdy(verbatimEventDate)) %>%
  filter(!is.na(date_common))
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'date_common = mdy(verbatimEventDate)'.
## Caused by warning:
## !  147 failed to parse.
```

```
final_df <- bee_df_ready %>%
  left_join(temp_df_agg, by = "date_common") %>%

  select(-any_of(c("date", "Date", "date_parsed", "verbatimEventDate",
                   "day2", "month2", "year2", "startDayofYear",
                   "endDayofYear", "source_file"))) %>%

  rename(date = date_common) %>%

  rename(
    temp_min_f        = tmin_degrees_f,
    temp_mean_f       = tmean_degrees_f,
    temp_max_f        = tmax_degrees_f,

    lat               = decimalLatitude,
    long              = decimalLongitude,
    gps_uncertainty   = coordinateUncertaintyInMeters,
    elevation         = verbatimElevation,
    state             = stateProvince,
    specimen_id       = specimenId,
    sample_id         = sampleId,
    sampling_method   = samplingProtocol
  )

print(colnames(final_df))
```

```
##  [1] "fieldNumber"           "catalogNumber"         "occurrenceID"
##  [4] "userId"                "userLogin"             "firstName"
##  [7] "firstNameInitial"      "lastName"              "recordedBy"
## [10] "sample_id"             "specimen_id"           "day"
## [13] "month"                 "year"                  "country"
## [16] "state"                 "county"                "locality"
## [19] "elevation"             "lat"                   "long"
## [22] "gps_uncertainty"       "sampling_method"       "relationshipOfResource"
## [25] "resourceID"            "relatedResourceID"     "relationshipRemarks"
## [28] "phylumPlant"           "orderPlant"            "familyPlant"
## [31] "genusPlant"            "speciesPlant"          "taxonRankPlant"
## [34] "url"                   "phylum"                "class"
## [37] "order"                 "family"                "genus"
## [40] "subgenus"              "specificEpithet"       "taxonomicNotes"
## [43] "scientificName"        "sex"                   "caste"
## [46] "taxonRank"             "identifiedBy"          "familyVolDet"
## [49] "genusVolDet"           "specificEpithetVolDet" "sexVolDet"
## [52] "casteVolDet"           "date"                  "temp_min_f"
## [55] "temp_mean_f"           "temp_max_f"            "vpdmin"
## [58] "vpdmax"
```

```r
print(head(final_df))
```

```
##            fieldNumber catalogNumber occurrenceID userId userLogin firstName
## 1 Jerry_Paul:18.007.001                             175529 jerrypaul     Jerry
## 2 Jerry_Paul:18.007.002                             175529 jerrypaul     Jerry
## 3 Jerry_Paul:18.007.003                             175529 jerrypaul     Jerry
## 4 Jerry_Paul:18.007.004                             175529 jerrypaul     Jerry
## 5 Jerry_Paul:18.007.005                             175529 jerrypaul     Jerry
## 6 Jerry_Paul:18.007.006                             175529 jerrypaul     Jerry
##   firstNameInitial lastName recordedBy sample_id specimen_id day month year
## 1               J.     Paul Jerry Paul         7           1  15     3 2017
## 2               J.     Paul Jerry Paul         7           2  15     3 2017
## 3               J.     Paul Jerry Paul         7           3  15     3 2017
## 4               J.     Paul Jerry Paul         7           4  15     3 2017
## 5               J.     Paul Jerry Paul         7           5  15     3 2017
## 6               J.     Paul Jerry Paul         7           6  15     3 2017
##   country state county                    locality elevation    lat     long
## 1     USA    OR Benton Corvallis, Highland Dell Dr       131 44.603 -123.264
## 2     USA    OR Benton Corvallis, Highland Dell Dr       131 44.603 -123.264
## 3     USA    OR Benton Corvallis, Highland Dell Dr       131 44.603 -123.264
## 4     USA    OR Benton Corvallis, Highland Dell Dr       131 44.603 -123.264
## 5     USA    OR Benton Corvallis, Highland Dell Dr       131 44.603 -123.264
## 6     USA    OR Benton Corvallis, Highland Dell Dr       131 44.603 -123.264
##   gps_uncertainty sampling_method relationshipOfResource resourceID
## 1              NA      aerial net                     NA
## 2              NA      aerial net                     NA
## 3              NA      aerial net                     NA
## 4              NA      aerial net                     NA
## 5              NA      aerial net                     NA
## 6              NA      aerial net                     NA
##   relatedResourceID relationshipRemarks phylumPlant orderPlant familyPlant
## 1                                  NA
## 2                                  NA
## 3                                  NA
## 4                                  NA
## 5                                  NA
## 6                                  NA
##   genusPlant speciesPlant taxonRankPlant url phylum class order family genus
## 1                                               NA    NA    NA     NA
## 2                                               NA    NA    NA     NA Osmia
## 3                                               NA    NA    NA     NA
## 4                                               NA    NA    NA     NA
## 5                                               NA    NA    NA     NA
## 6                                               NA    NA    NA     NA
##   subgenus specificEpithet taxonomicNotes scientificName  sex caste taxonRank
## 1                                                                          NA
## 2                 lignaria                               male              NA
## 3                                                                          NA
## 4                                                                          NA
## 5                                                                          NA
## 6                                                                          NA
##   identifiedBy familyVolDet genusVolDet specificEpithetVolDet sexVolDet
## 1
```

```
## 2     L.R.Best
## 3
## 4
## 5
## 6
##   casteVolDet       date temp_min_f temp_mean_f temp_max_f vpdmin vpdmax
## 1             2017-03-15         NA          NA         NA     NA     NA
## 2             2017-03-15         NA          NA         NA     NA     NA
## 3             2017-03-15         NA          NA         NA     NA     NA
## 4             2017-03-15         NA          NA         NA     NA     NA
## 5             2017-03-15         NA          NA         NA     NA     NA
## 6             2017-03-15         NA          NA         NA     NA     NA
```

```
final_df %>%
  count(genus, sort = TRUE)
```

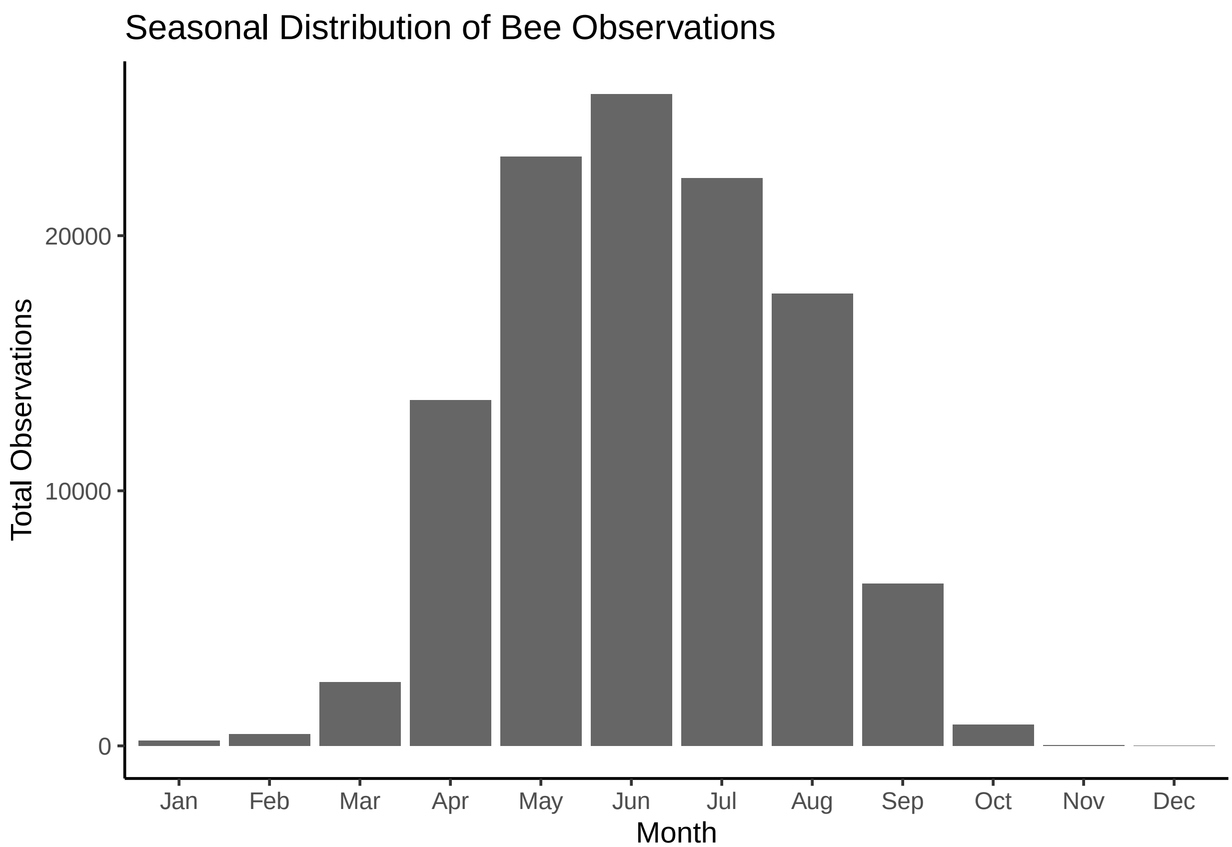
```
##              genus     n
## 1                  25499
## 2     Lasioglossum 12577
## 3           Bombus 10581
## 4            Osmia  9041
## 5         Halictus  8200
## 6          Andrena  6788
## 7         Ceratina  5603
## 8        Megachile  3602
## 9       Melissodes  3349
## 10         Hylaeus  2836
## 11      Anthophora  2194
## 12          Nomada  1906
## 13        Andrena   1813
## 14       Halictus   1810
## 15        Colletes  1702
## 16     Agapostemon  1407
## 17      Panurginus  1221
## 18         Perdita  1221
## 19        Hoplitis  1069
## 20          Eucera   974
## 21            Apis   953
## 22       Sphecodes   828
## 23    Ashmeadiella   766
## 24       Anthidium   716
## 25     Dianthidium   609
## 26    Agapostemon    565
## 27       Protosmia   460
## 28        Diadasia   447
## 29        Heriades   408
## 30   Brachymelecta   376
## 31        Dufourea   350
## 32       Habropoda   314
## 33       Coelioxys   304
## 34      Triepeolus   287
## 35          Osmia    235
## 36          Stelis   195
## 37      Chelostoma   191
```

```
## 38       Atoposmia   190
## 39      Anthidium    167
## 40    Anthidiellum   159
## 41      Calliopsis   144
## 42        Xylocopa   114
## 43         Epeolus    79
## 44         Melecta    68
## 45    Anthophorula    39
## 46        Neolarra    27
## 47         Biastes    25
## 48     Protandrena    25
## 49        Trachusa    24
## 50           Nomia    23
## 51       Hoplitis     14
## 52   Tetraloniella    11
## 53          Dioxys    10
## 54 Pseudoanthidium    10
## 55     Oreopasites     9
## 56   Xenoglossodes     7
## 57  Micralictoides     5
## 58        Zacosmia     5
## 59   Lasioglossum      4
## 60       Peponapis     4
## 61   Epimelissodes     3
## 62      Habropoda      3
## 63      Megachile      3
## 64       Colletes      2
## 65   Ashmeadiella      1
```

```r
seasonality_plot <- final_df %>%
  mutate(month = lubridate::month(date, label = TRUE)) %>%
  count(month, native_status) %>%
  ggplot(aes(x = month, y = n, fill = native_status)) +
  geom_col() +
  labs(
    title = "Data Check 1: Seasonal Distribution of Observations",
    x = "Month", y = "Count of Records"
  ) +
  scale_fill_manual(values = c("Native" = "#00BFC4", "Introduced" = "#F8766D")) +
  theme_minimal()

print(seasonality_plot)
```

```r
seasonality_plot <- final_df %>%
  mutate(month_label = lubridate::month(date, label = TRUE)) %>%
  count(month_label) %>%
  ggplot(aes(x = month_label, y = n)) +

  geom_col(fill = "gray40") +

  labs(
    title = "Seasonal Distribution of Bee Observations",
    x = "Month",
    y = "Total Observations"
```

```r
  ) +
  theme_classic()

print(seasonality_plot)
```




```r
species_list <- final_df %>%
  mutate(scientific_name = paste(genus, specificEpithet, sep = " ")) %>%
  distinct(scientific_name) %>%
  arrange(scientific_name) %>%
  pull(scientific_name)

print(species_list)
```

```
##   [1] " "                            "Agapostemon  femoratus"
##   [3] "Agapostemon  subtilior"       "Agapostemon  virescens "
##   [5] "Agapostemon femoratus"        "Agapostemon melliventris"
##   [7] "Agapostemon subtilior"        "Agapostemon virescens"
##   [9] "Andrena "                     "Andrena  "
##  [11] "Andrena amphibola"            "Andrena angustitarsata"
##  [13] "Andrena anisochlora"          "Andrena annectens"
##  [15] "Andrena astragali"            "Andrena barbilabris"
##  [17] "Andrena bendensis"            "Andrena birtwelli"
##  [19] "Andrena candida"              "Andrena candidiformis"
##  [21] "Andrena cerasifolii"          "Andrena chapmanae"
##  [23] "Andrena chlorogaster"         "Andrena cleodora"
```

```
##  [25] "Andrena colletina"             "Andrena columbiana"
##  [27] "Andrena commoda"               "Andrena crataegi"
##  [29] "Andrena cressonii"             "Andrena cupreotincta"
##  [31] "Andrena cyanophila"            "Andrena erythrogaster"
##  [33] "Andrena fuscicauda"            "Andrena hippotes"
##  [35] "Andrena hypoleuca"             "Andrena illinoiensis"
##  [37] "Andrena lawrencei"             "Andrena medionitens"
##  [39] "Andrena melanochroa"           "Andrena merriami"
##  [41] "Andrena microchlora"           "Andrena miranda"
##  [43] "Andrena miserabilis"           "Andrena nigrocaerulea"
##  [45] "Andrena nivalis"               "Andrena orthocarpi"
##  [47] "Andrena pallidifovea"          "Andrena pensilis"
##  [49] "Andrena perarmata"             "Andrena perplexa"
##  [51] "Andrena piperi"                "Andrena prunorum"
##  [53] "Andrena quintiliformis"        "Andrena saliciforis"
##  [55] "Andrena scurra"                "Andrena scutellinitens"
##  [57] "Andrena semipunctata"          "Andrena striatifrons"
##  [59] "Andrena subaustralis"          "Andrena sulcata"
##  [61] "Andrena surda"                 "Andrena topazana"
##  [63] "Andrena transnigra"            "Andrena trevoris"
##  [65] "Andrena vicina"                "Andrena w-scripta"
##  [67] "Andrena walleyi"               "Andrena wheeleri"
##  [69] "Anthidiellum robertsoni"       "Anthidium "
##  [71] "Anthidium  "                   "Anthidium  banningense"
##  [73] "Anthidium  formosum"           "Anthidium  illustre"
##  [75] "Anthidium  palliventre"        "Anthidium atrifrons"
##  [77] "Anthidium atripes"             "Anthidium banningense"
##  [79] "Anthidium clypeodentatum"      "Anthidium edwardsii"
##  [81] "Anthidium emarginatum"         "Anthidium formosum"
##  [83] "Anthidium illustre"            "Anthidium maculosum"
##  [85] "Anthidium manicatum"           "Anthidium mormonum"
##  [87] "Anthidium oblongatum"          "Anthidium palliventre"
##  [89] "Anthidium placitum"            "Anthidium tenuiflorae"
##  [91] "Anthidium utahense"            "Anthophora "
##  [93] "Anthophora affabilis"          "Anthophora albata"
##  [95] "Anthophora bomboides"          "Anthophora californica"
##  [97] "Anthophora curta"              "Anthophora edwardsii"
##  [99] "Anthophora exigua"             "Anthophora maculifrons"
## [101] "Anthophora pacifica"           "Anthophora peritomae"
## [103] "Anthophora porterae"           "Anthophora terminalis"
## [105] "Anthophora urbana"             "Anthophora ursina"
## [107] "Anthophorula "                 "Anthophorula chionura"
## [109] "Apis mellifera"                "Ashmeadiella "
## [111] "Ashmeadiella  difugita"        "Ashmeadiella altadenae"
## [113] "Ashmeadiella aridula"          "Ashmeadiella bucconis"
## [115] "Ashmeadiella cactorum"         "Ashmeadiella californica"
## [117] "Ashmeadiella cubiceps"         "Ashmeadiella difugita"
## [119] "Ashmeadiella foveata"          "Ashmeadiella foxiella"
## [121] "Ashmeadiella meliloti"         "Ashmeadiella prosopidis"
## [123] "Ashmeadiella timberlakei"      "Atoposmia "
## [125] "Atoposmia abjecta"             "Atoposmia anthodyta"
## [127] "Atoposmia copelandica"         "Atoposmia elongata"
## [129] "Atoposmia oregona"             "Atoposmia triodonta"
## [131] "Biastes "                      "Biastes fulviventris"
```

```
## [133] "Bombus "                    "Bombus appositus"
## [135] "Bombus caliginosus"         "Bombus centralis"
## [137] "Bombus fervidus"            "Bombus flavidus"
## [139] "Bombus flavifrons"          "Bombus griseocollis"
## [141] "Bombus huntii"              "Bombus impatiens"
## [143] "Bombus insularis"           "Bombus melanopygus"
## [145] "Bombus mixtus"              "Bombus morrisoni"
## [147] "Bombus nevadensis"          "Bombus occidentalis"
## [149] "Bombus rufocinctus"         "Bombus sitkensis"
## [151] "Bombus sylvicola"           "Bombus vagans"
## [153] "Bombus vancouverensis"      "Bombus vandykei"
## [155] "Bombus vosnesenskii"        "Brachymelecta californica"
## [157] "Calliopsis "                "Calliopsis anthidia"
## [159] "Calliopsis coloradensis"    "Calliopsis edwardsii"
## [161] "Calliopsis fracta"          "Calliopsis michenerella"
## [163] "Calliopsis obscurella"      "Calliopsis personata"
## [165] "Calliopsis puellae"         "Calliopsis scitula"
## [167] "Calliopsis scutellaris"     "Calliopsis xenus"
## [169] "Ceratina "                  "Ceratina acantha"
## [171] "Ceratina acantha "          "Ceratina arizonensis"
## [173] "Ceratina dallatorreana"     "Ceratina hurdi"
## [175] "Ceratina micheneri"         "Ceratina nanula"
## [177] "Ceratina neomexicana"       "Ceratina pacifica"
## [179] "Ceratina sequoiae"          "Ceratina tejonensis"
## [181] "Chelostoma "                "Chelostoma californicum"
## [183] "Chelostoma minutum"         "Chelostoma phaceliae"
## [185] "Coelioxys "                 "Coelioxys alternatus"
## [187] "Coelioxys apacheorum"       "Coelioxys banksi"
## [189] "Coelioxys deani"            "Coelioxys edita"
## [191] "Coelioxys erysimi"          "Coelioxys grindeliae"
## [193] "Coelioxys hirsutissimus"    "Coelioxys moestus"
## [195] "Coelioxys novomexicanus"    "Coelioxys octodentatus"
## [197] "Coelioxys porterae"         "Coelioxys rufitarsis"
## [199] "Coelioxys serricaudatus"    "Coelioxys sodalis"
## [201] "Colletes "                  "Colletes  "
## [203] "Colletes compactus"         "Colletes consors"
## [205] "Colletes fulgidus"          "Colletes gypsicolens"
## [207] "Colletes kincaidii"         "Colletes ligatus"
## [209] "Colletes nigrifrons"        "Colletes phaceliae"
## [211] "Colletes simulans"          "Colletes slevini"
## [213] "Diadasia "                  "Diadasia angusticeps"
## [215] "Diadasia australis"         "Diadasia bituberculata"
## [217] "Diadasia diminuta"          "Diadasia enavata"
## [219] "Diadasia lutzi"             "Diadasia nigrifrons"
## [221] "Diadasia nitidifrons"       "Dianthidium "
## [223] "Dianthidium curvatum"       "Dianthidium dubium"
## [225] "Dianthidium heterulkei"     "Dianthidium parvum"
## [227] "Dianthidium platyurum"      "Dianthidium plenum"
## [229] "Dianthidium pudicum"        "Dianthidium singulare"
## [231] "Dianthidium subparvum"      "Dianthidium ulkei"
## [233] "Dioxys "                    "Dioxys aurifuscus"
## [235] "Dioxys pacificus"           "Dioxys pomonae"
## [237] "Dufourea "                  "Dufourea holocyanea"
## [239] "Dufourea trochantera"       "Dufourea versatilis"
```

```
## [241] "Epeolus "                   "Epeolus compactus"
## [243] "Epeolus novomexicanus"       "Epeolus olympiellus"
## [245] "Epimelissodes obliqua"       "Eucera "
## [247] "Eucera acerba"               "Eucera actuosa"
## [249] "Eucera amsinckiae"           "Eucera cordleyi"
## [251] "Eucera edwardsii"            "Eucera frater"
## [253] "Eucera fulvitarsis"          "Eucera hurdi"
## [255] "Eucera lunata"               "Eucera speciosa"
## [257] "Eucera virgata"              "Habropoda "
## [259] "Habropoda  "                 "Habropoda cineraria"
## [261] "Habropoda dammersi"          "Habropoda depressa"
## [263] "Habropoda miserabilis"       "Habropoda tristissima"
## [265] "Halictus "                   "Halictus  confusus"
## [267] "Halictus  confusus "         "Halictus  farinosus"
## [269] "Halictus  ligatus"           "Halictus  rubicundus"
## [271] "Halictus  tripartitus"       "Halictus  tripartitus "
## [273] "Halictus confusus"           "Halictus farinosus"
## [275] "Halictus farinosus "         "Halictus ligatus"
## [277] "Halictus rubicundus"         "Halictus tripartitus"
## [279] "Halictus virgatellus"        "Heriades "
## [281] "Heriades carinata"           "Heriades cressoni"
## [283] "Heriades variolosa"          "Hoplitis "
## [285] "Hoplitis  "                  "Hoplitis albifrons"
## [287] "Hoplitis boharti"            "Hoplitis emarginata"
## [289] "Hoplitis fulgida"            "Hoplitis grinnelli"
## [291] "Hoplitis hypocrita"          "Hoplitis louisae"
## [293] "Hoplitis orthognatha"        "Hoplitis plagiostoma"
## [295] "Hoplitis producta"           "Hoplitis remotula"
## [297] "Hoplitis robusta"            "Hoplitis sambuci"
## [299] "Hoplitis uvulalis"           "Hoplitis viridimicans"
## [301] "Hylaeus "                    "Hylaeus Mesillae"
## [303] "Hylaeus basalis"             "Hylaeus mesillae"
## [305] "Hylaeus nunenmacheri"        "Lasioglossum "
## [307] "Lasioglossum  "              "Lasioglossum  anhypops"
## [309] "Lasioglossum  helianthi"     "Lasioglossum  trizonatum "
## [311] "Lasioglossum aberrans"       "Lasioglossum albipenne"
## [313] "Lasioglossum albohirtum"     "Lasioglossum allonotus"
## [315] "Lasioglossum anhypops"       "Lasioglossum argemonis"
## [317] "Lasioglossum aspilurum"      "Lasioglossum boreale"
## [319] "Lasioglossum brunneiventre"  "Lasioglossum buccale"
## [321] "Lasioglossum cembrilacus"    "Lasioglossum colatum"
## [323] "Lasioglossum cooleyi"        "Lasioglossum cordleyi"
## [325] "Lasioglossum cressonii"      "Lasioglossum dashwoodi"
## [327] "Lasioglossum diatretum"      "Lasioglossum diversopunctatum"
## [329] "Lasioglossum ebmerellum"     "Lasioglossum egregium"
## [331] "Lasioglossum glabriventre"   "Lasioglossum helianthi"
## [333] "Lasioglossum hudsoniellum"   "Lasioglossum hyalinum"
## [335] "Lasioglossum incompletum"    "Lasioglossum incompletum "
## [337] "Lasioglossum inconditum"     "Lasioglossum kincaidii"
## [339] "Lasioglossum knereri"        "Lasioglossum laevissimum"
## [341] "Lasioglossum lusorium"       "Lasioglossum macoupinense"
## [343] "Lasioglossum macroprosopum"  "Lasioglossum marinense"
## [345] "Lasioglossum megastictus"    "Lasioglossum mellipes"
## [347] "Lasioglossum nevadense"      "Lasioglossum nigrum"
```

```
## [349] "Lasioglossum novascotiae"      "Lasioglossum occultum"
## [351] "Lasioglossum occulturn"        "Lasioglossum olympiae"
## [353] "Lasioglossum orthocarpi"       "Lasioglossum ovaliceps"
## [355] "Lasioglossum pacatum"          "Lasioglossum pacificum"
## [357] "Lasioglossum pacificum "       "Lasioglossum pavonotus"
## [359] "Lasioglossum prasinogaster"    "Lasioglossum pruinosum"
## [361] "Lasioglossum pulveris"         "Lasioglossum punctatoventre"
## [363] "Lasioglossum quebecense"       "Lasioglossum robustum"
## [365] "Lasioglossum ruidosense"       "Lasioglossum sandhousiellum"
## [367] "Lasioglossum sedi"             "Lasioglossum sequoiae"
## [369] "Lasioglossum sisymbrii"        "Lasioglossum tegulariforme"
## [371] "Lasioglossum titusi"           "Lasioglossum trizonatum"
## [373] "Lasioglossum tuolumnense"      "Lasioglossum villosulum"
## [375] "Lasioglossum yukonae"          "Lasioglossum zephyrus"
## [377] "Lasioglossum zonulus"          "Megachile "
## [379] "Megachile  perihirta"          "Megachile addenda"
## [381] "Megachile angelarum"           "Megachile anograe"
## [383] "Megachile apicalis"            "Megachile brevis"
## [385] "Megachile centuncularis"       "Megachile cleomis"
## [387] "Megachile coquilletti"         "Megachile fidelis"
## [389] "Megachile frigida"             "Megachile gemula"
## [391] "Megachile gentilis"            "Megachile gravita"
## [393] "Megachile inermis"             "Megachile inimica"
## [395] "Megachile lapponica"           "Megachile lippiae"
## [397] "Megachile melanophaea"         "Megachile mellitarsis"
## [399] "Megachile montivaga"           "Megachile nevadensis"
## [401] "Megachile onobrychidis"        "Megachile parallela"
## [403] "Megachile perihirta"           "Megachile pseudonigra"
## [405] "Megachile pugnata"             "Megachile relativa"
## [407] "Megachile rotundata"           "Megachile subnigra"
## [409] "Megachile texana"              "Megachile umatillensis"
## [411] "Megachile wheeleri"            "Melecta "
## [413] "Melecta edwardsii"             "Melecta pacifica"
## [415] "Melecta separata"              "Melecta thoracica"
## [417] "Melissodes "                   "Melissodes agilis"
## [419] "Melissodes bimatris"           "Melissodes clarkiae"
## [421] "Melissodes communis"           "Melissodes lupinus"
## [423] "Melissodes lustrus"            "Melissodes metenuus"
## [425] "Melissodes microstictus"       "Melissodes pallidisignatus"
## [427] "Melissodes rivalis"            "Melissodes robustior"
## [429] "Melissodes semilupinus"        "Micralictoides ruficaudus"
## [431] "Neolarra "                     "Neolarra vandykei"
## [433] "Neolarra vigilans"             "Nomada "
## [435] "Nomia melanderi"               "Oreopasites "
## [437] "Oreopasites vanduzeei"         "Osmia "
## [439] "Osmia  "                       "Osmia  densa"
## [441] "Osmia  kincaidii"              "Osmia  lignaria"
## [443] "Osmia  longula"                "Osmia  montana"
## [445] "Osmia  nemoris"                "Osmia  obliqua"
## [447] "Osmia aglaia"                  "Osmia albolateralis"
## [449] "Osmia atriventris"             "Osmia atrocyanea"
## [451] "Osmia bella"                   "Osmia brevis"
## [453] "Osmia bruneri"                 "Osmia bucephala"
## [455] "Osmia caerulescens"            "Osmia californica"
```

```
## [457] "Osmia calla"                    "Osmia cara"
## [459] "Osmia caraformis"               "Osmia cobaltina"
## [461] "Osmia coloradensis"             "Osmia cornifrons"
## [463] "Osmia cyanella"                 "Osmia densa"
## [465] "Osmia dolerosa"                 "Osmia ednae"
## [467] "Osmia enixa"                    "Osmia exigua"
## [469] "Osmia gabrielis"                "Osmia gaudiosa"
## [471] "Osmia giliarum"                 "Osmia glauca"
## [473] "Osmia grinnelli"                "Osmia inermis"
## [475] "Osmia integra"                  "Osmia inurbana"
## [477] "Osmia juxta"                    "Osmia kincaidii"
## [479] "Osmia laeta"                    "Osmia lanei"
## [481] "Osmia lignaria"                 "Osmia longula"
## [483] "Osmia malina"                   "Osmia marginipennis"
## [485] "Osmia melanopleura"             "Osmia montana"
## [487] "Osmia nanula"                   "Osmia nemoris"
## [489] "Osmia nifoata"                  "Osmia nigrifrons"
## [491] "Osmia obliqua"                  "Osmia odontogaster"
## [493] "Osmia paradisica"               "Osmia pentstemonis"
## [495] "Osmia pikei"                    "Osmia proxima"
## [497] "Osmia pusilla"                  "Osmia raritatis"
## [499] "Osmia rawlinsi"                 "Osmia regulina"
## [501] "Osmia ribifloris"               "Osmia sanrafaelae"
## [503] "Osmia sculleni"                 "Osmia simillima"
## [505] "Osmia subaustralis"             "Osmia tanneri"
## [507] "Osmia tarsata"                  "Osmia tersula"
## [509] "Osmia texana"                   "Osmia thysanisca"
## [511] "Osmia tristella"                "Osmia unca"
## [513] "Osmia vandykei"                 "Panurginus "
## [515] "Peponapis pruinosa"             "Perdita "
## [517] "Perdita albipennis"             "Perdita nevadensis"
## [519] "Protandrena "                   "Protosmia rubifloris"
## [521] "Pseudoanthidium nanum"          "Sphecodes "
## [523] "Sphecodes arvensiformis"        "Sphecodes pecosensis"
## [525] "Stelis "                        "Stelis ashmeadiellae"
## [527] "Stelis carnifex"                "Stelis laticincta"
## [529] "Stelis monticola"               "Stelis rubi"
## [531] "Stelis subemarginata"           "Tetraloniella pomonae"
## [533] "Trachusa timberlakei"           "Triepeolus "
## [535] "Triepeolus argyreus"            "Triepeolus californicus"
## [537] "Triepeolus concavus"            "Triepeolus helianthi"
## [539] "Triepeolus lunatus"             "Triepeolus melanarius"
## [541] "Triepeolus paenepectoralis"     "Triepeolus utahensis"
## [543] "Triepeolus verbesinae complex" "Xenoglossodes "
## [545] "Xylocopa californica"           "Xylocopa tabaniformis"
## [547] "Xylocopa virginica"             "Zacosmia maculata"
```

```
final_df <- final_df %>%
  mutate(
    genus = str_squish(genus),
    specificEpithet = str_squish(specificEpithet),
    scientific_name = paste(genus, specificEpithet)
  ) %>%
```

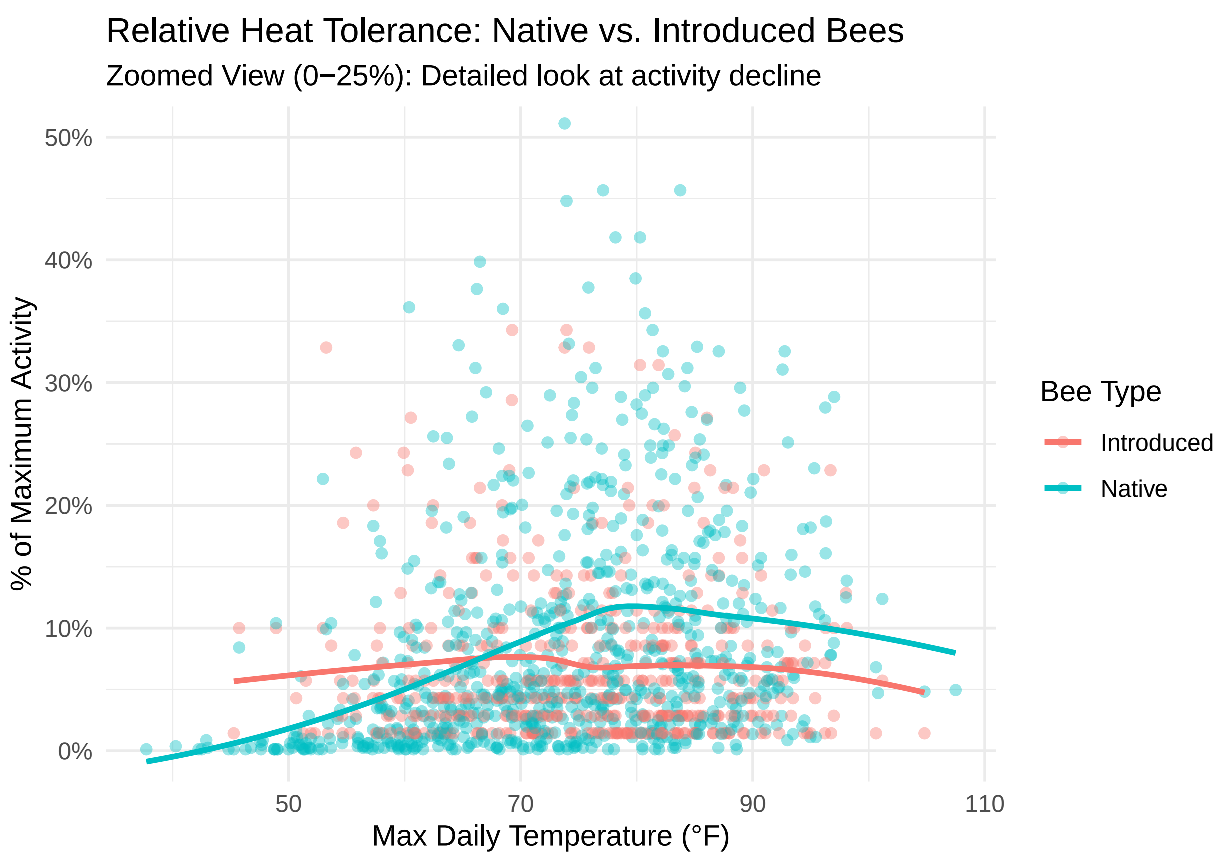
```r
  filter(genus != "", specificEpithet != "") %>%

  mutate(
    native_status = case_when(
      genus == "Apis" ~ "Introduced",
      genus == "Anthidium" & specificEpithet == "manicatum" ~ "Introduced",
      genus == "Megachile" & specificEpithet == "rotundata" ~ "Introduced",
      genus == "Osmia" & specificEpithet == "cornifrons" ~ "Introduced",
      genus == "Osmia" & specificEpithet == "caerulescens" ~ "Introduced",
      genus == "Lasioglossum" & specificEpithet == "zonulus" ~ "Introduced",
      genus == "Lasioglossum" & specificEpithet == "villosulum" ~ "Introduced",
      genus == "Xylocopa" & specificEpithet == "virginica" ~ "Introduced",

      TRUE ~ "Native"
    )
  )

table(final_df$native_status)
```

```
##
## Introduced     Native
##       2533      55486
```

```r
daily_summary <- final_df %>%
  group_by(date, temp_max_f, native_status) %>%
  summarize(
    total_bees = n(),
    species_count = n_distinct(scientific_name),
    .groups = "drop"
  )

normalized_data <- daily_summary %>%
  group_by(native_status) %>%
  mutate(
    max_observed = max(total_bees),

    percent_activity = (total_bees / max_observed) * 100
  ) %>%
  ungroup()

ggplot(normalized_data, aes(x = temp_max_f, y = percent_activity, color = native_status)) +

  geom_point(alpha = 0.4, position = "jitter") +

  geom_smooth(method = "loess", se = FALSE) +

  coord_cartesian(ylim = c(0, 50)) +

  labs(
    title = "Relative Heat Tolerance: Native vs. Introduced Bees",
    subtitle = "Zoomed View (0-25%): Detailed look at activity decline",
    x = "Max Daily Temperature (°F)",
    y = "% of Maximum Activity",
```

```r
    color = "Bee Type"
  ) +
  scale_y_continuous(labels = function(x) paste0(x, "%")) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 85 rows containing non-finite values (`stat_smooth()`).

## Warning: Removed 85 rows containing missing values (`geom_point()`).
```




```r
summer_stats_data <- daily_summary %>%
  mutate(month = month(date)) %>%
  filter(month >= 5 & month <= 9)

natives_only <- summer_stats_data %>% filter(native_status == "Native")
introduced_only <- summer_stats_data %>% filter(native_status == "Introduced")

test_native <- cor.test(natives_only$total_bees,
                        natives_only$temp_max_f,
                        method = "spearman")
```

```
## Warning in cor.test.default(natives_only$total_bees, natives_only$temp_max_f, :
## Cannot compute exact p-value with ties
```

```r
test_introduced <- cor.test(introduced_only$total_bees,
                            introduced_only$temp_max_f,
                            method = "spearman")
```

```
## Warning in cor.test.default(introduced_only$total_bees,
## introduced_only$temp_max_f, : Cannot compute exact p-value with ties
```

```r
print("--- NATIVE BEES RESULTS ---")
```

```
## [1] "--- NATIVE BEES RESULTS ---"
```

```r
print(test_native)
```

```
##
##  Spearman's rank correlation rho
##
## data:  natives_only$total_bees and natives_only$temp_max_f
## S = 27469379, p-value = 0.0009337
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.1375499
```

```r
print("--- INTRODUCED BEES RESULTS ---")
```

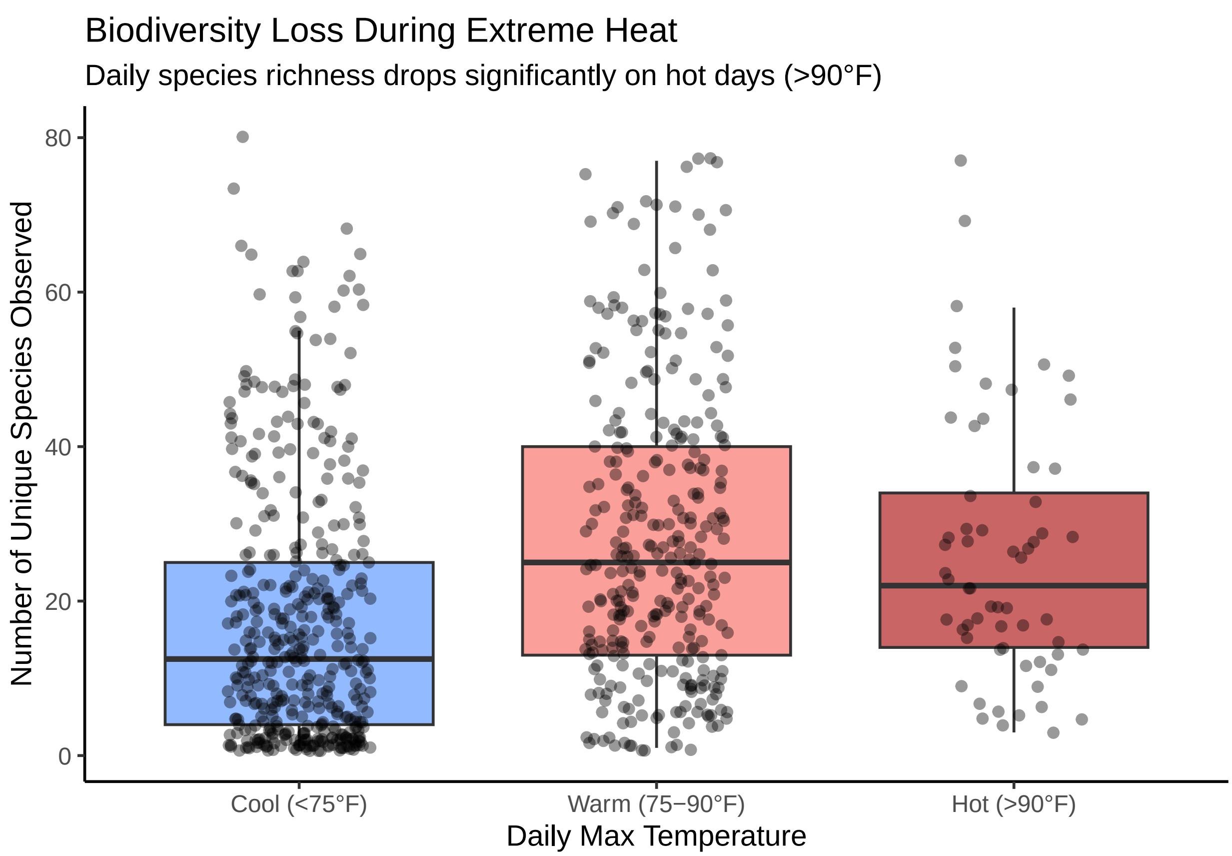
```
## [1] "--- INTRODUCED BEES RESULTS ---"
```

```r
print(test_introduced)
```

```
##
##  Spearman's rank correlation rho
##
## data:  introduced_only$total_bees and introduced_only$temp_max_f
## S = 10902485, p-value = 0.6507
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## 0.02253483
```

```r
richness_data <- final_df %>%
  filter(!is.na(temp_max_f)) %>%

  mutate(temp_category = cut(temp_max_f,
                             breaks = c(-Inf, 75, 90, Inf),
                             labels = c("Cool (<75°F)", "Warm (75-90°F)", "Hot (>90°F)"))) %>%

  filter(!is.na(temp_category)) %>%

  group_by(date, temp_category) %>%
```

```
  summarize(daily_richness = n_distinct(scientific_name), .groups = "drop")

ggplot(richness_data, aes(x = temp_category, y = daily_richness, fill = temp_category)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.4, color = "black") +
  labs(
    title = "Biodiversity Loss During Extreme Heat",
    subtitle = "Daily species richness drops significantly on hot days (>90°F)",
    x = "Daily Max Temperature",
    y = "Number of Unique Species Observed"
  ) +
  theme_classic() +
  scale_fill_manual(values = c("Cool (<75°F)" = "#619CFF",
                               "Warm (75-90°F)" = "#F8766D",
                               "Hot (>90°F)" = "#B22222")) +
  theme(legend.position = "none")
```




```
split_richness_data <- final_df %>%
  mutate(month = month(date)) %>%
  filter(month >= 5 & month <= 9) %>%

  filter(!is.na(temp_max_f)) %>%

  mutate(temp_category = cut(temp_max_f,
                             breaks = c(-Inf, 75, 90, Inf),
```

```
                              labels = c("Cool (<75°F)", "Warm (75-90°F)", "Hot (>90°F)"))) %>%

  filter(!is.na(temp_category)) %>%

  group_by(date, temp_category, native_status) %>%
  summarize(daily_richness = n_distinct(scientific_name), .groups = "drop")

native_richness <- split_richness_data %>% filter(native_status == "Native")
kw_native <- kruskal.test(daily_richness ~ temp_category, data = native_richness)

introduced_richness <- split_richness_data %>% filter(native_status == "Introduced")
kw_introduced <- kruskal.test(daily_richness ~ temp_category, data = introduced_richness)

print("--- NATIVE RICHNESS (Kruskal-Wallis) ---")
```

```
## [1] "--- NATIVE RICHNESS (Kruskal-Wallis) ---"
```

```
print(kw_native)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  daily_richness by temp_category
## Kruskal-Wallis chi-squared = 7.7403, df = 2, p-value = 0.02086
```

```
print("--- INTRODUCED RICHNESS (Kruskal-Wallis) ---")
```

```
## [1] "--- INTRODUCED RICHNESS (Kruskal-Wallis) ---"
```

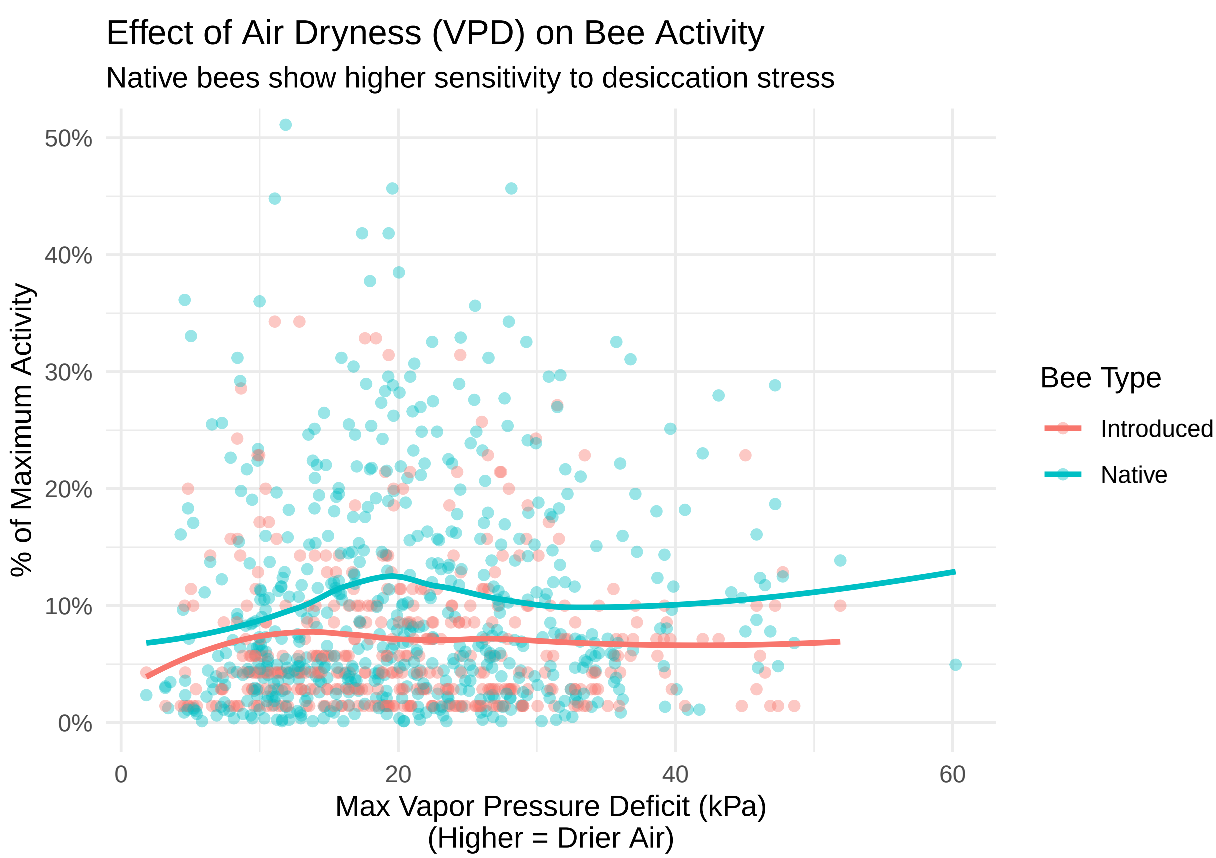
```
print(kw_introduced)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  daily_richness by temp_category
## Kruskal-Wallis chi-squared = 1.0949, df = 2, p-value = 0.5784
```

```
vpd_plot_data <- final_df %>%
  mutate(month = month(date)) %>%
  filter(month >= 5 & month <= 9) %>%

  group_by(date, vpdmax, native_status) %>%
  summarize(total_bees = n(), .groups = "drop") %>%

  filter(!is.na(vpdmax)) %>%

  group_by(native_status) %>%
  mutate(
    max_observed = max(total_bees),
    percent_activity = (total_bees / max_observed) * 100
```

```
  ) %>%
  ungroup()

ggplot(vpd_plot_data, aes(x = vpdmax, y = percent_activity, color = native_status)) +

  geom_point(alpha = 0.4, position = "jitter") +

  geom_smooth(method = "loess", se = FALSE) +

  coord_cartesian(ylim = c(0, 50)) +

  labs(
    title = "Effect of Air Dryness (VPD) on Bee Activity",
    subtitle = "Native bees show higher sensitivity to desiccation stress",
    x = "Max Vapor Pressure Deficit (kPa)\n(Higher = Drier Air)",
    y = "% of Maximum Activity",
    color = "Bee Type"
  ) +
  scale_y_continuous(labels = function(x) paste0(x, "%")) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```r
native_vpd <- vpd_plot_data %>% filter(native_status == "Native")
intro_vpd <- vpd_plot_data %>% filter(native_status == "Introduced")

test_native_vpd <- cor.test(native_vpd$total_bees, native_vpd$vpdmax, method = "spearman")
```

```
## Warning in cor.test.default(native_vpd$total_bees, native_vpd$vpdmax, method =
## "spearman"): Cannot compute exact p-value with ties
```

```r
test_intro_vpd <- cor.test(intro_vpd$total_bees, intro_vpd$vpdmax, method = "spearman")
```

```
## Warning in cor.test.default(intro_vpd$total_bees, intro_vpd$vpdmax, method =
## "spearman"): Cannot compute exact p-value with ties
```

```r
print("--- NATIVE VPD RESULTS ---")
```

```
## [1] "--- NATIVE VPD RESULTS ---"
```

```r
print(test_native_vpd)
```

```
##
##  Spearman's rank correlation rho
##
## data:  native_vpd$total_bees and native_vpd$vpdmax
## S = 27951459, p-value = 0.003254
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.1224142
```

```r
print("--- INTRODUCED VPD RESULTS ---")
```

```
## [1] "--- INTRODUCED VPD RESULTS ---"
```

```r
print(test_intro_vpd)
```

```
##
##  Spearman's rank correlation rho
##
## data:  intro_vpd$total_bees and intro_vpd$vpdmax
## S = 10870054, p-value = 0.6092
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## 0.02544243
```

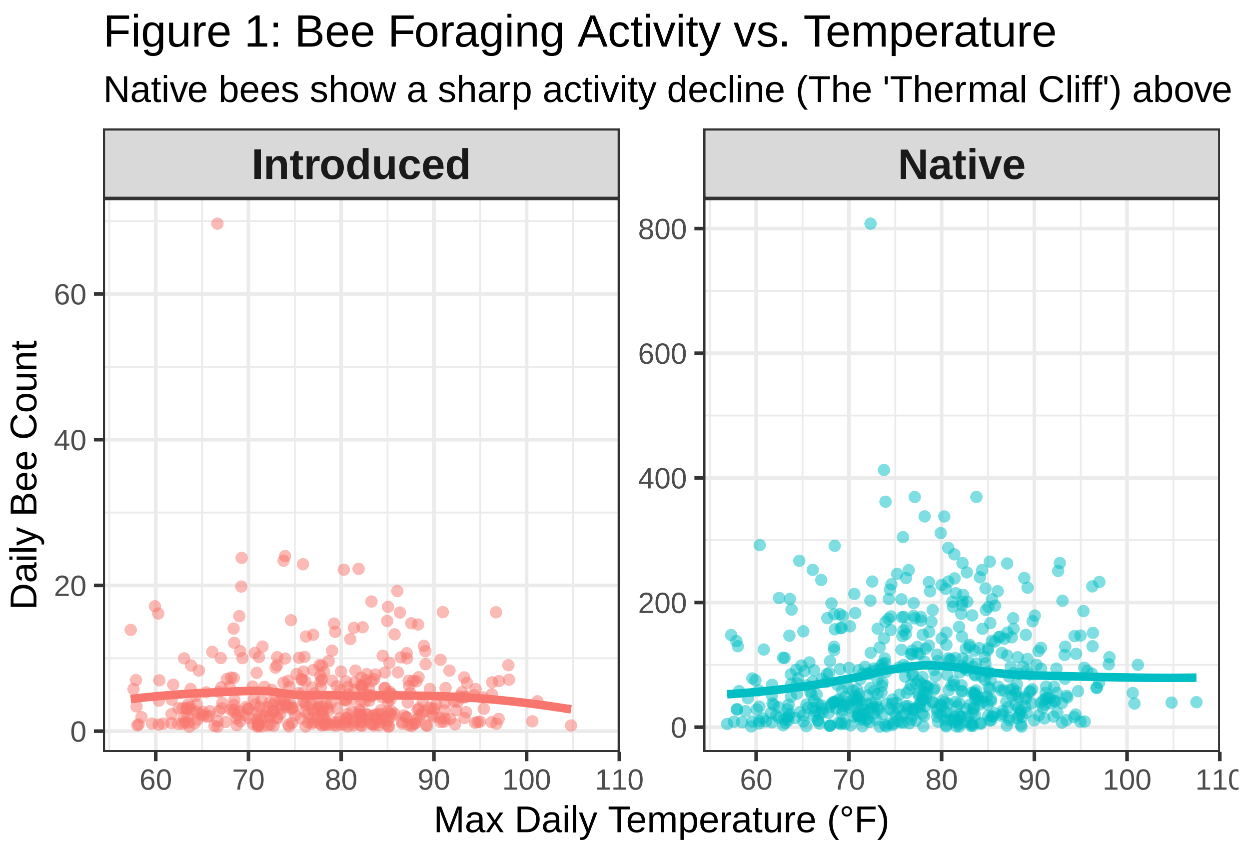
```r
summer_df <- final_df %>%
  mutate(month = month(date)) %>%
  filter(month >= 5 & month <= 9) %>%
```

```r
  filter(!is.na(temp_max_f))

bee_colors <- c("Native" = "#00BFC4", "Introduced" = "#F8766D")

plot1_data <- summer_df %>%
  group_by(date, temp_max_f, native_status) %>%
  summarize(total_bees = n(), .groups = "drop")

plot1 <- ggplot(plot1_data, aes(x = temp_max_f, y = total_bees, color = native_status)) +
  geom_point(alpha = 0.5, position = "jitter") +
  geom_smooth(method = "loess", se = FALSE, size = 1.5) +
  facet_wrap(~native_status, scales = "free_y") +
  labs(
    title = "Figure 1: Bee Foraging Activity vs. Temperature",
    subtitle = "Native bees show a sharp activity decline (The 'Thermal Cliff') above 90°F",
    x = "Max Daily Temperature (°F)",
    y = "Daily Bee Count",
    color = "Bee Type"
  ) +
  scale_color_manual(values = bee_colors) +
  theme_bw(base_size = 14) +
  theme(legend.position = "none", strip.text = element_text(face = "bold", size = 16))
```

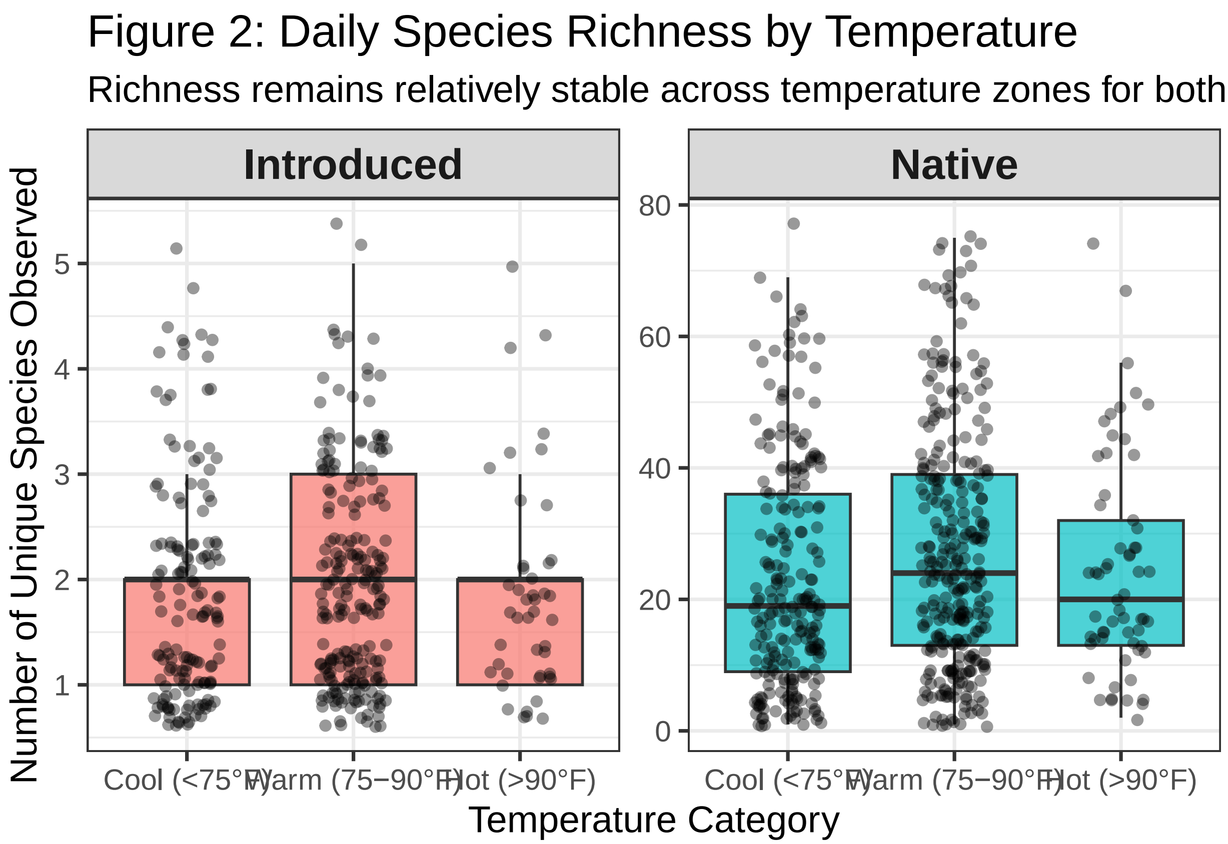
```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```r
print(plot1)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
plot2_data <- summer_df %>%
  mutate(temp_category = cut(temp_max_f,
                             breaks = c(-Inf, 75, 90, Inf),
                             labels = c("Cool (<75°F)", "Warm (75-90°F)", "Hot (>90°F)"))) %>%
  filter(!is.na(temp_category)) %>%
  group_by(date, temp_category, native_status) %>%
  summarize(daily_richness = n_distinct(scientific_name), .groups = "drop")

plot2 <- ggplot(plot2_data, aes(x = temp_category, y = daily_richness, fill = native_status)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.4) +
  facet_wrap(~native_status, scales = "free_y") +
  labs(
    title = "Figure 2: Daily Species Richness by Temperature",
    subtitle = "Richness remains relatively stable across temperature zones for both groups",
    x = "Temperature Category",
    y = "Number of Unique Species Observed"
  ) +
  scale_fill_manual(values = bee_colors) +
  theme_bw(base_size = 14) +
  theme(legend.position = "none", strip.text = element_text(face = "bold", size = 16))

print(plot2)
```

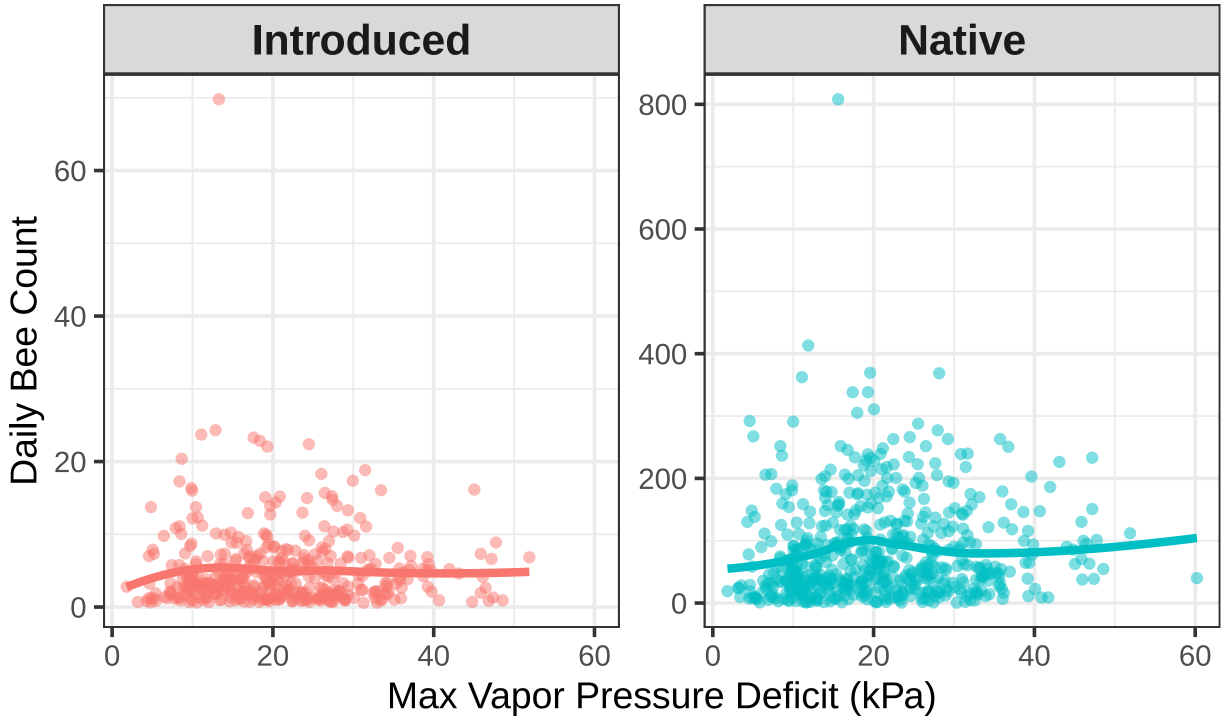



```
plot3_data <- summer_df %>%
  filter(!is.na(vpdmax)) %>%
  group_by(date, vpdmax, native_status) %>%
  summarize(total_bees = n(), .groups = "drop")

plot3 <- ggplot(plot3_data, aes(x = vpdmax, y = total_bees, color = native_status)) +
  geom_point(alpha = 0.5, position = "jitter") +
  geom_smooth(method = "loess", se = FALSE, size = 1.5) +
  facet_wrap(~native_status, scales = "free_y") +
  labs(
    title = "Figure 3: Impact of Air Dryness (VPD) on Activity",
    subtitle = "Native activity drops as air becomes drier (Higher VPD)",
    x = "Max Vapor Pressure Deficit (kPa)",
    y = "Daily Bee Count",
    color = "Bee Type"
  ) +
  scale_color_manual(values = bee_colors) +
  theme_bw(base_size = 14) +
  theme(legend.position = "none", strip.text = element_text(face = "bold", size = 16))

print(plot3)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Figure 3: Impact of Air Dryness (VPD) on Activity

Native activity drops as air becomes drier (Higher VPD)




```r
print("--- STATS FOR FIGURE 1 (Abundance vs Temp) ---")
```

```
## [1] "--- STATS FOR FIGURE 1 (Abundance vs Temp) ---"
```

```r
# Test: Spearman Correlation
p1_native <- plot1_data %>% filter(native_status == "Native")
p1_intro  <- plot1_data %>% filter(native_status == "Introduced")

res1_native <- cor.test(p1_native$total_bees, p1_native$temp_max_f, method = "spearman")
```

```
## Warning in cor.test.default(p1_native$total_bees, p1_native$temp_max_f, :
## Cannot compute exact p-value with ties
```

```r
res1_intro  <- cor.test(p1_intro$total_bees, p1_intro$temp_max_f, method = "spearman")
```

```
## Warning in cor.test.default(p1_intro$total_bees, p1_intro$temp_max_f, method =
## "spearman"): Cannot compute exact p-value with ties
```

```r
print("Native Abundance Correlation:")
```

```
## [1] "Native Abundance Correlation:"
```

```r
print(res1_native)
```

```
##
##  Spearman's rank correlation rho
##
## data:  p1_native$total_bees and p1_native$temp_max_f
## S = 27469379, p-value = 0.0009337
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.1375499
```

```r
print("Introduced Abundance Correlation:")
```

```
## [1] "Introduced Abundance Correlation:"
```

```r
print(res1_intro)
```

```
##
##  Spearman's rank correlation rho
##
## data:  p1_intro$total_bees and p1_intro$temp_max_f
## S = 10902485, p-value = 0.6507
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## 0.02253483
```

```r
print("--- STATS FOR FIGURE 2 (Richness vs Temp Category) ---")
```

```
## [1] "--- STATS FOR FIGURE 2 (Richness vs Temp Category) ---"
```

```r
# Test: Kruskal-Wallis Rank Sum Test
p2_native <- plot2_data %>% filter(native_status == "Native")
p2_intro  <- plot2_data %>% filter(native_status == "Introduced")

res2_native <- kruskal.test(daily_richness ~ temp_category, data = p2_native)
res2_intro  <- kruskal.test(daily_richness ~ temp_category, data = p2_intro)

print("Native Richness K-W Test:")
```

```
## [1] "Native Richness K-W Test:"
```

```r
print(res2_native)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  daily_richness by temp_category
## Kruskal-Wallis chi-squared = 7.7403, df = 2, p-value = 0.02086
```

```r
print("Introduced Richness K-W Test:")
```

```
## [1] "Introduced Richness K-W Test:"
```

```r
print(res2_intro)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  daily_richness by temp_category
## Kruskal-Wallis chi-squared = 1.0949, df = 2, p-value = 0.5784
```

```r
print("--- STATS FOR FIGURE 3 (Abundance vs VPD) ---")
```

```
## [1] "--- STATS FOR FIGURE 3 (Abundance vs VPD) ---"
```

```r
# Test: Spearman Correlation
p3_native <- plot3_data %>% filter(native_status == "Native")
p3_intro  <- plot3_data %>% filter(native_status == "Introduced")

res3_native <- cor.test(p3_native$total_bees, p3_native$vpdmax, method = "spearman")
```

```
## Warning in cor.test.default(p3_native$total_bees, p3_native$vpdmax, method =
## "spearman"): Cannot compute exact p-value with ties
```

```r
res3_intro  <- cor.test(p3_intro$total_bees, p3_intro$vpdmax, method = "spearman")
```

```
## Warning in cor.test.default(p3_intro$total_bees, p3_intro$vpdmax, method =
## "spearman"): Cannot compute exact p-value with ties
```

```r
print("Native VPD Correlation:")
```

```
## [1] "Native VPD Correlation:"
```

```r
print(res3_native)
```

```
##
##  Spearman's rank correlation rho
##
## data:  p3_native$total_bees and p3_native$vpdmax
## S = 27951459, p-value = 0.003254
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.1224142
```

```r
print("Introduced VPD Correlation:")
```

```
## [1] "Introduced VPD Correlation:"
```

```r
print(res3_intro)
```

```
##
##  Spearman's rank correlation rho
##
## data:  p3_intro$total_bees and p3_intro$vpdmax
## S = 10870054, p-value = 0.6092
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## 0.02544243
```

Conclusion

Our analysis of the Oregon Bee Atlas and PRISM climate data reveals a physiological divergence in how native and introduced bees respond to the environmental conditions.

Answering our primary question, we found that native bees are significantly more sensitive to environmental effects than introduced species. Native foraging abundance showed a highly significant correlation with maximum daily temperature ($p < 0.001$). In contrast, introduced species dominated by the European Honey Bee showed no statistical relationship with temperature ($p = 0.65$). This indicates that introduced generalists possess a "thermal resilience" allowing them to exploit resources consistently across a wide range of temperatures.

Unlike our initial predictions of stability, the statistical test revealed that native species richness varies significantly across temperature categories ($p = 0.02$), while introduced richness remains static ($p = 0.58$). This suggests that extreme heat does not just reduce the number of native bees flying, but actively filters the types of native bees present, likely excluding species with narrower thermal windows from the landscape during heat waves.

Our results confirmed the hypothesis that air dryness acts as a limiting factor for native communities. Native abundance was significantly correlated with Vapor Pressure Deficit ($p = 0.003$), while introduced species showed no sensitivity to aridity ($p = 0.60$). This supports the theory that smaller bodied native bees are more susceptible to VPD stress than managed species, putting them at distinct disadvantage as climate change leads to drier summers in the Pacific Northwest.

These findings provide strong evidence that introduced honey bees have a competitive "stability advantage" in a warming climate. Conservation strategies that rely on honey bee data to assess "pollinator health" will mask the stress being experienced by native communities. To prevent the loss of native biodiversity, land managers must prioritize the protection of shade providing wildlife and diverse canopy structures to buffer the effects of extreme heat and aridity on sensitive native populations.

References

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